

SAI JAHNAVI DAMACHARLA

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EDUCATION

Indiana University, *Luddy School of Informatics, Computing & Engineering, Indianapolis, IN*

Aug 2024- Present

Master of Science in Applied Data Science

Course Work: Data Science Programming, Machine Learning, Computer Vision, Database Design, Statistics with R, Cloud Computing.

Acharya Nagarjuna University, *Bapatla Engineering College, Bapatla, India*

Jan 2020 - May 2024

Bachelor of Technology in Computer Science & Engineering

Course Work: Python, Computer Networks, Operating Systems, Data Structures and Algorithms, DevOps, Web Development, Full stack development(MERN).

TECHNICAL SKILLS

- **Programming Languages**: Python (Pandas, NumPy, ScikitLearn etc.), R, SQL
- **Tools and IDE**: Power BI, Tableau, Jupyter Notebook, Google Colab, MS Access, RStudio, VSCode, PyCharm
- **Data Analysis**: Data Cleaning and Preprocessing, EDA and Visualization, Statistical Modelling, Feature Engineering
- **Geospatial Tools**: ArcGIS Pro, ArcGIS Experience Builder, ESRI, Leaflet.js, Google Earth Engine, Rasterio, PyProj
- **Web & Tools**: Vue.js, Bootstrap, Leaflet.js, GitHub, Git version control, Docker, Kubernetes, HTML, CSS, JS
- **Frameworks/Cloud**: CI/CD pipelines, GCP, ArcGIS Pro, ArcGIS Experience Builder, ESRI, Google Earth Engine

WORK EXPERIENCE

GIS Analyst Summer Intern, *Christopher B. Burke Engineering, LLC. (CBBEL), Indianapolis, IN*

May 2025 – Aug 2025

- Worked with **ArcGIS Pro** to develop visual exhibits and perform advanced geospatial analysis for environmental planning and civil engineering projects in the counties of Indiana.
- Participated in meetings, contributing to discussions and providing spatial insights to support decision-making.
- Explored innovative approaches to integrate Deep Learning and Advanced Visualization techniques into GIS workflows, aiming to enhance the efficiency and effectiveness of the planning department's operations.

Graduate Assistant, *Luddy School of Informatics, Indiana University, Indianapolis, IN*

January 2025 - May 2025

- Collaborated with professor to deliver lectures, overseen labs, and facilitated discussions in computer science.
- Provided academic support, clarifying doubts in algorithms and computer networking subject concepts for students.
- Engaged in faculty-led research and technical documentation proof reading for the needed corrections in the work.

PROJECTS

Deep Learning for Building Footprint Extraction – GIS & GeoAI Project

May 2025 – Aug 2025

- Setup Deep Learning and **GeoAI** environments and extensions in **ArcGIS Pro**, integrating GIS-specific Python libraries.
- Explored spatial data sources including **USGS imagery** and **.shp files** to support geospatial model development.
- Fetched and fine-tuned a building footprint extraction model with a city using deep learning tools with esri.
- Conducted geo-segmentation analysis, implemented **3D modeling** features using ESRI's 3D Features training.
- Utilized **Google Colab** for collaborative AI experimentation and ran model performance evaluations.
- Participated in discussions and webinars on floodplain connectivity, AI automation, and **MHMP data** harvesting with professionals from IDEM and ESRI.
- Contributed to the MHMP website outline, supporting emergency planning with spatial insights.
- Engaged in **AI tools evaluation** and contributed to selecting research-appropriate tools for the GeoAI team.

Spotify Music Tracks Analysis Project using R Language

- Explored Spotify tracks dataset to identify trends in popularity, artist collaborations, genre distribution, and audio features like valence and energy.
- Utilized dplyr, tidyverse, and ggplot2 for data cleaning, aggregation, and visualization through bar charts, scatter plots, and trend lines.

Digital Forensics & Cyber Security Threat Management Database Project

- Designed and implemented a relational database system to track cyber threats, digital evidence, incident reports, and analyst activities using MSAccess and SQL queries.
- Built sql queries and forms to enable efficient data retrieval, evidence linking, and incident tracking for forensic data.

EXTRACURRICULARS

Professional Development Lead, *Data Science & Machine Learning Club, IUI*

October 2024 - Present

- Working as a career-leader from the club of DSML, designed and facilitated workshops and sessions to guide aspiring data scientists in building impactful portfolios.

PUBLICATION WORK

- Published a research paper in JETIR on Soil-Moisture based Motor Pumping for Agriculture

August 2022